

ECG-Noise-Classfier

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Overview

This project focuses on ECG Signal Quality Assessment by classifying different types of noise artifacts using Deep Learning techniques.

The ECG signals and noise sources were obtained from the MIT-BIH Noise Stress Test Database using the WFDB library. The system is designed to identify whether an ECG segment is clean or corrupted by specific types of interference.

The system performs:

- Signal Acquisition: Loading raw ECG and noise records.
- Preprocessing: DC-offset removal and Z-score normalization.
- Temporal Windowing: Segmenting signals into fixed 300-sample windows.
- Noise Classification: Using a 1D Convolutional Neural Network (CNN) to distinguish between:
 - Clean Signals
 - Baseline Wander (BW)
 - Muscle Artifacts (EM)

- Motion Artifacts (MA)
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Software Tools

- Language: Python
 - Environment: PyCharm / Jupyter Notebook
 - Key Libraries:
 - wfdb: For medical database access.
 - numpy & pandas: For data manipulation.
 - matplotlib & seaborn: For signal and performance visualization.
 - scikit-learn: For data splitting and metrics.
 - tensorflow / keras: For building and training the CNN.
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Dataset Information

- Source: MIT-BIH Noise Stress Test Database.
 - Records Used: 118e00 through 119e12 for clean signals and bw, em, ma for noise profiles.
 - Sampling: All signals are processed at their native sampling frequency (360 Hz) and segmented into windows of 300 samples.
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Software Implementation

1. Signal Loading & Segmentation

Signals are fetched via the wfdb package.

- Clean Data: Extracted from patient records using a sliding window with 50% overlap.
- Noise Data: Loaded from specific noise records and segmented similarly to create a balanced dataset.

2. Signal Preprocessing (Normalization)

To ensure the model focuses on the "shape" of the noise rather than its amplitude:

- Mean Removal: Each segment is centered at zero.
- Standardization: Dividing by the standard deviation (Z-score) to normalize energy across all samples.

3. Model Architecture (1D-CNN)

The core of the project is a deep 1D-CNN designed for high stability:

- Feature Extraction: Three Convolutional layers (Conv1D) with increasing filters (32, 64, 128).
- Normalization: BatchNormalization after each convolution to speed up convergence.
- Global Average Pooling: Used instead of Flattening to reduce parameters and prevent overfitting to specific temporal locations.
- Regularization: Dropout (0.5) and EarlyStopping with ReduceLROnPlateau for adaptive learning.

Results & Evaluation

- Test Accuracy: Achieved ~94% overall accuracy.
- Class Performance:
 - BW & MA: Exceptional detection rates (>97% F1-score).
 - EM: Identified as the most challenging class due to its high-frequency overlap with clean ECG components.
- Visual Outputs:
 - Stabilized Training/Validation Accuracy & Loss curves.
 - Detailed Confusion Matrix showing inter-class correlations.
 - Power Spectral Density (PSD) plots for frequency analysis.

Conclusion

- **Successfully implemented an automated pipeline for ECG quality assessment.**
 - **Demonstrated that Global Average Pooling significantly stabilizes validation performance in 1D signal tasks.**
 - **The model effectively differentiates between low-frequency wander and high-frequency muscle noise, providing a reliable preprocessing tool for cardiac monitors.**
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Future Work

- **Integrating LSTM layers to capture longer temporal dependencies.**
- **Deploying the model using TensorFlow Lite for real-time edge device monitoring.**
- **Testing the model on different databases (e.g., PhysioNet Challenge) to ensure generalizability.**